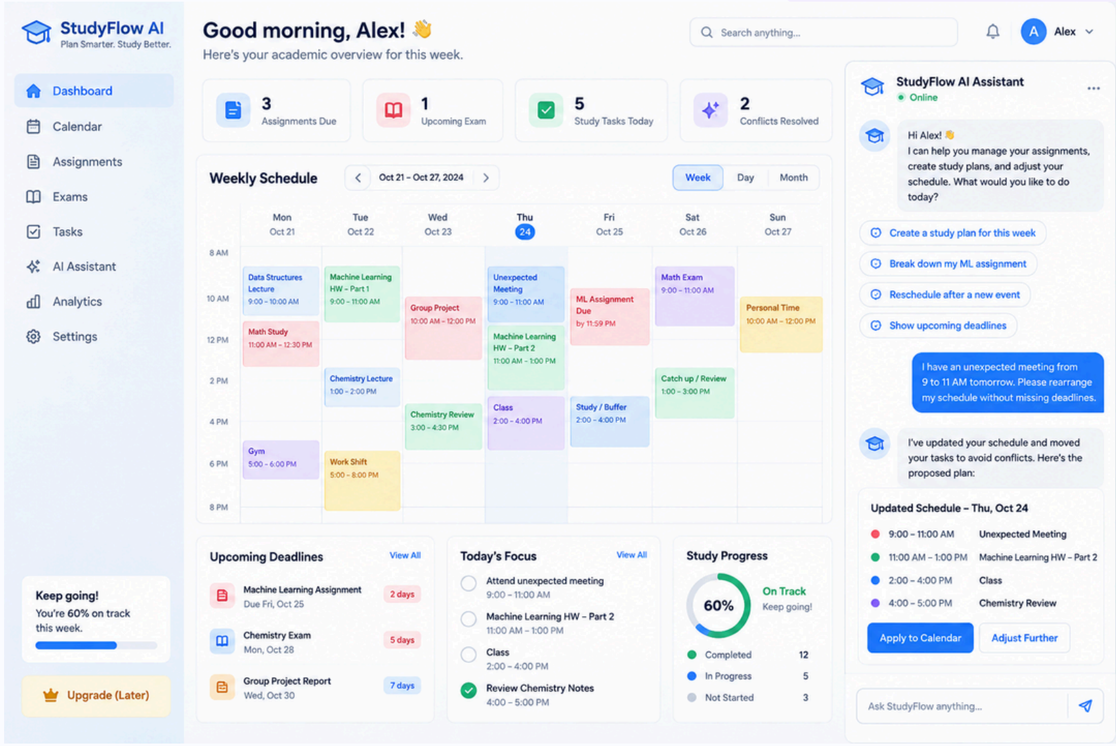
Agentic AI Hackathon Project

# StudyFlow AI

An adaptive academic planning assistant that helps students organize assignments, generate study plans, and recover when schedules change.

Assignments | Tasks | Scheduling | Replanning




**Plan smarter. Study better.**

Team StudyFlow AI

THE PROBLEM

# Students do not just need reminders. They need a plan that stays useful when life changes.

> *“ I know what is due. I just do not know when to do it, how long it will take, or how to recover when my day gets disrupted. ”*

This is the real academic planning problem we want to solve.

WHY IT MATTERS

# Academic planning breaks in three predictable ways.

### Fragmented deadlines

Assignments, exams, classes, and personal commitments live across different tools and become hard to manage.

### Planning overhead

Students must estimate effort, break down work, and choose priorities before they even begin studying.

### Unexpected disruptions

A missed study block or a new meeting can make a carefully built plan obsolete.

**1**

problem space

**A planner that cannot adapt creates more stress instead of reducing it.**

THE SOLUTION

# StudyFlow AI turns academic responsibilities into an executable weekly plan.

Students add their assignments, exams, and fixed commitments. StudyFlow AI breaks work into smaller tasks, creates study blocks, checks feasibility, and replans when something changes.

**The output**

A practical weekly plan
not just an AI to-do list.

WORKFLOW

# From academic chaos to a clear plan in three steps.

## 1. Capture

Add assignments, exams, deadlines, classes, and other fixed events.

## 2. Plan

Generate tasks, estimate effort, and create study blocks around available time.

## 3. Adapt

If the week changes, revise the schedule and ask the student for approval.

WHY AGENTIC AI

# A static scheduler can organize time. An agentic system can reason, react, and recover.

## Traditional planner

- Stores deadlines
- Creates reminders
- May generate a schedule once
- Breaks when the student misses time
- Leaves replanning to the user

## StudyFlow AI

- Interprets natural-language requests
- Breaks assignments into tasks
- Uses scheduling tools and validation
- Detects conflicts and repairs plans
- Requests approval before changes

Core idea: intelligence is not just in generating a plan — it is in deciding what to do next when that plan fails.

PRODUCT SURFACES

# StudyFlow AI is designed as a complete academic planning workspace.

## Dashboard

Weekly overview, schedule, deadlines, and AI assistant.

## Assignments

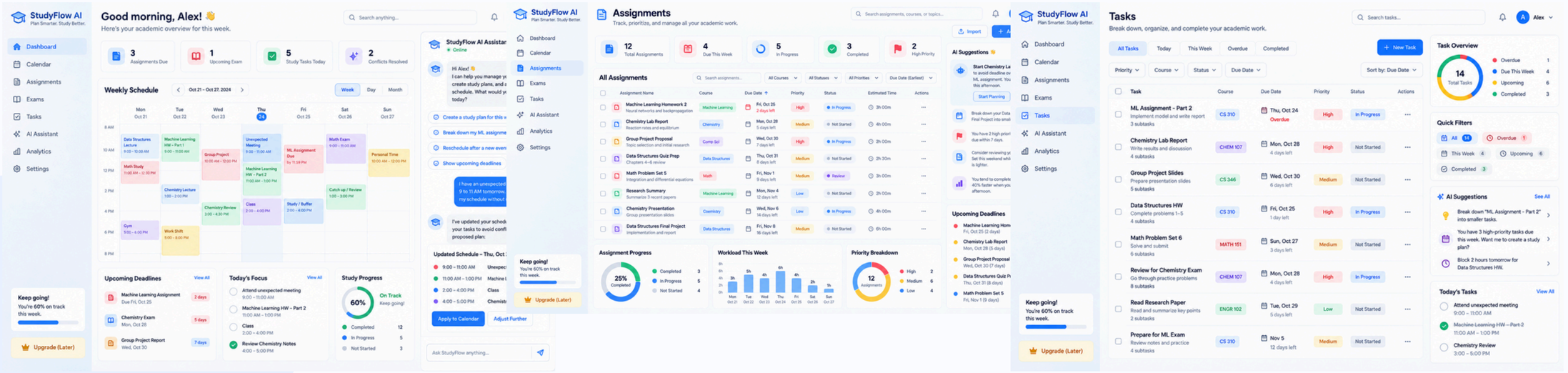
Track due dates, priorities, status, and estimated effort.

## Tasks

Manage subtasks, progress, and focused study actions.

## Calendar + AI

Generate plans and replan after disruptions.



UI CONCEPT

# A clean student-facing experience focused on clarity, action, and confidence.




## Dashboard

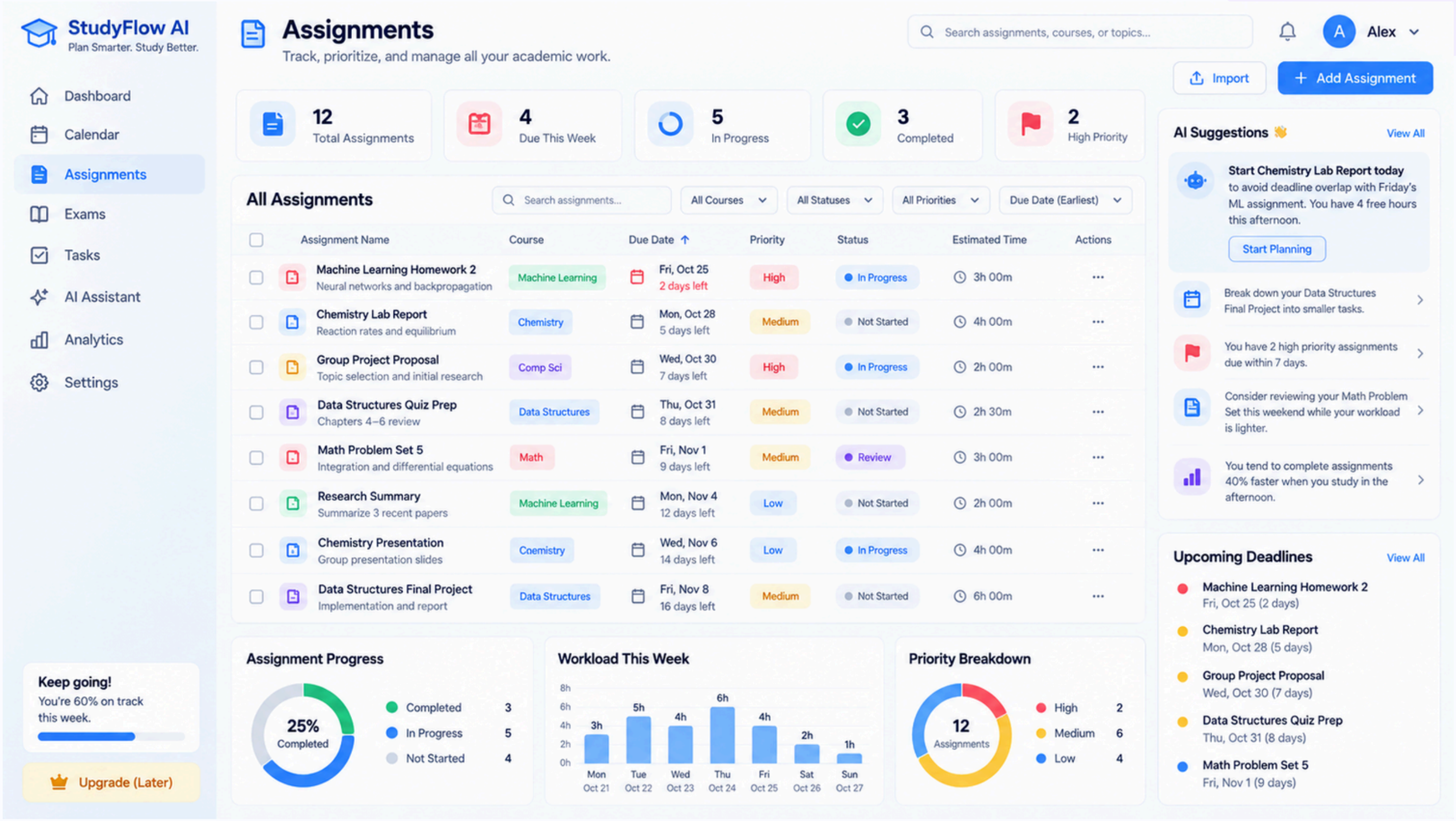
Weekly calendar, focus tasks, progress, and AI guidance.

PRODUCT SURFACES

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## Assignments

Track due dates, priorities, status, and estimated effort.

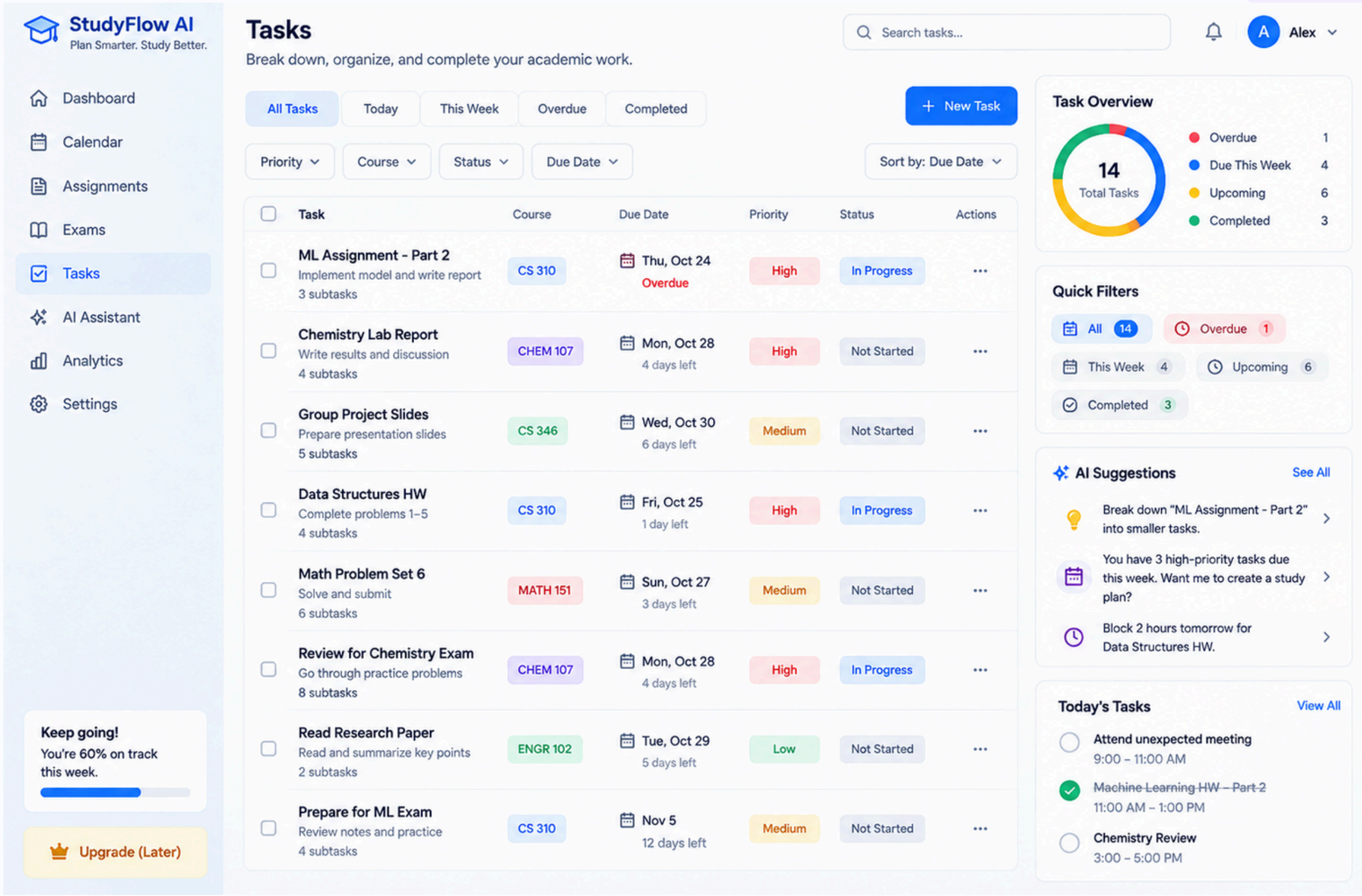


PRODUCT SURFACES

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## Tasks

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LIVE DEMO STORY

# The strongest moment in the demo is when the schedule repairs itself.

Student says: “I have an unexpected meeting tomorrow from 9 AM to 11 AM. Rearrange my schedule without missing deadlines.”

## Before

9–11 AM  ML homework
11–12 PM  Chemistry review
1–3 PM  Class (fixed)
3–4 PM  Available
4–5 PM  Available

## After

9–11 AM  Unexpected meeting
11–1 PM  ML homework – part 2
1–3 PM  Class (fixed)
3–4 PM  Chemistry review
4–5 PM  Study buffer

Result: the plan adapts, constraints are preserved, and the student stays in control through approval.

SYSTEM ARCHITECTURE

# StudyFlow AI combines agents, deterministic scheduling logic, and a student-facing web app.

## Frontend

React + TypeScript + Tailwind UI for Dashboard, Assignments, Tasks, Calendar, and AI Assistant

### Supervisor

Routes requests

### Task Planning

Breaks down work

### Scheduling

Allocates study time

### Replanning

Repairs schedules

### Validation

Checks feasibility

## Backend + Data

FastAPI • LangGraph • SQLite • scheduling tools • approval state • activity logs

WHAT MAKES STUDYFLOW SPECIAL?

# It is not just a planner. It is a planning system that can explain and repair decisions.

## End-to-end flow

Assignment → task → calendar → replanning

## Human-in-the-loop

All meaningful schedule edits are reviewed before they are applied

## Observable AI

Students can see agent activity, proposed changes, and reasons

FUTURE VISION

# StudyFlow AI can grow into a truly personalized academic operating system.

## Near-term extensions

Google Calendar sync
Canvas integration
Reminders and notifications

## Smarter personalization

Preferred study times
Task-duration learning
Adaptive workload balancing

## Longer-term direction

Performance insights
Semester planning
More advanced student support agents

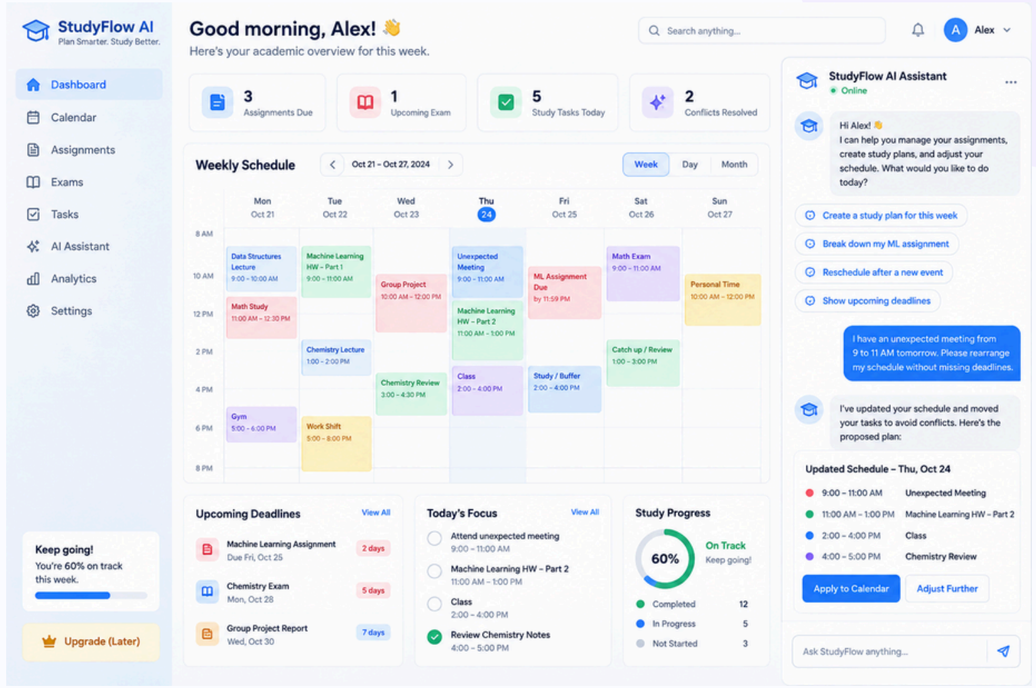
**Vision: help students spend less time planning and more time actually learning.**

Thank you

# StudyFlow AI

## Questions?

Helping students turn deadlines into realistic action.




Paul English Applied AI Institute • UMass Boston

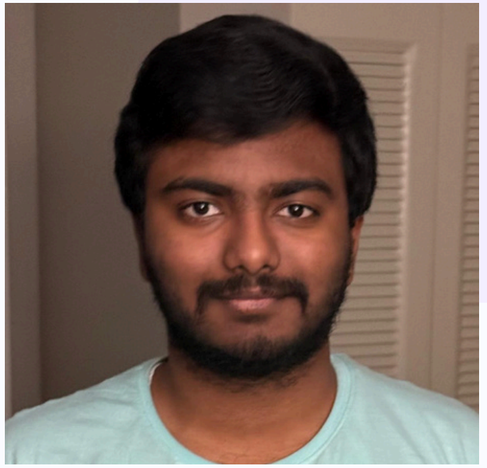
# Team Kanyarashi




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